

Q1a_i

$$\text{average thickness} = 3.8 / 20 = 0.19 \text{ cm}$$

Q1b

$$\text{density} = \text{mass} / \text{volume} = 148 / 16.6 = 8.9 \text{ g/cm}^3$$

Q4a

$$\text{speed} = \text{distance} / \text{time} = 200 / 6.4 = 31 \text{ m/s}$$

Q5b

$$\begin{aligned} \text{clockwise moments} &= \text{anticlockwise moments. } 200 = (2.0 \times 10) + (F \times 60). F = \\ (200 - 20) / 60 &= 180 / 60 = 3.0 \text{ N} \end{aligned}$$

Q11b

$$V_s/V_p = N_s/N_p, \text{ so } V_s = (64 / 240) \times 960 = 16 \text{ V}$$

Q12b

$$137 - 56 = 81$$

Q12c

$$\text{three half-lives have passed, so } 36 / 8 = 4.5 \text{ mg}$$